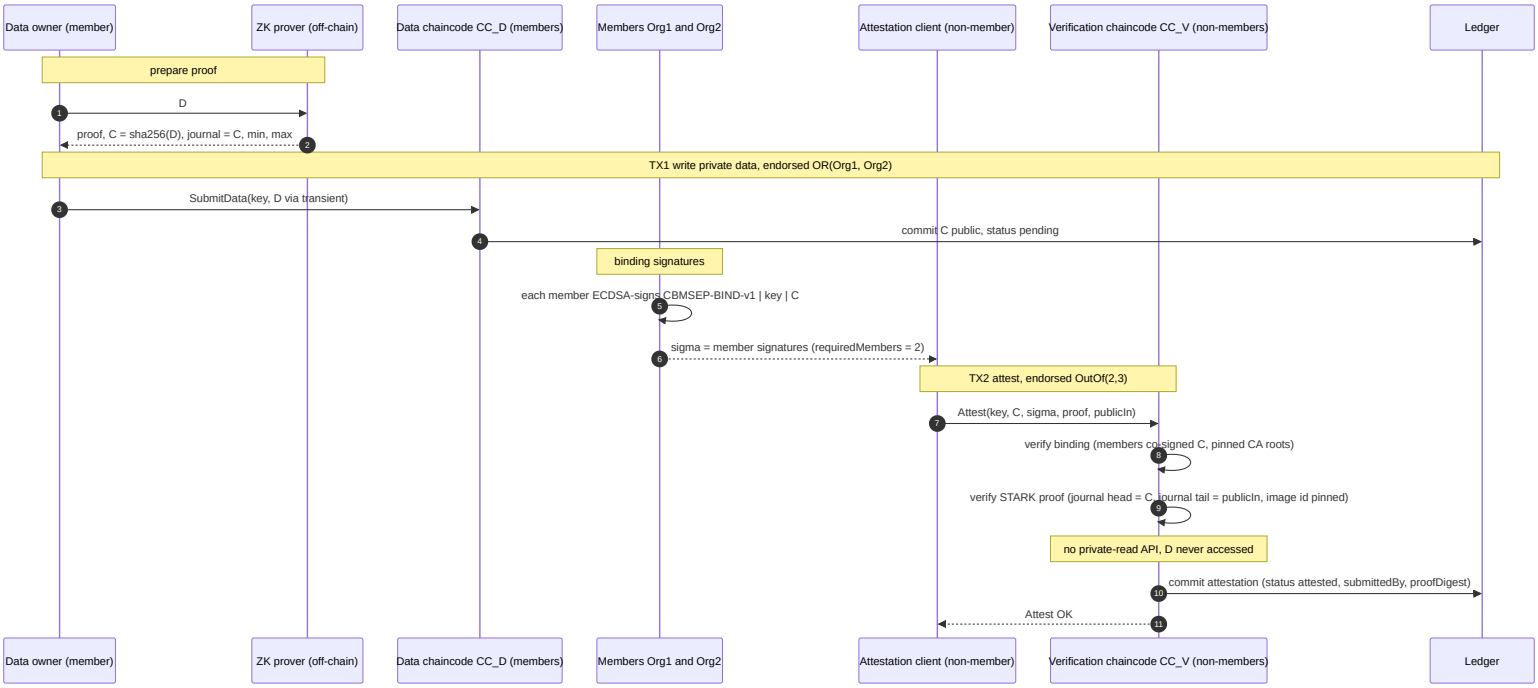
Data owner (member)
ZK prover (off-chain)
Data chaincode CC_D (members)
Members Org1 and Org2
Attestation client (non-member)
Verification chaincode CC_V (non-members)
Ledger
prepare proof
1
D
2
proof, C = sha256(D), journal = C, min, max
TX1 write private data, endorsed OR(Org1, Org2)
3
SubmitData(key, D via transient)
4
commit C public, status pending
binding signatures
5
each member ECDSA-signs CBMSEP-BIND-v1 | key | C
6
sigma = member signatures (requiredMembers = 2)
TX2 attest, endorsed OutOf(2,3)
7
Attest(key, C, sigma, proof, publicIn)
8
verify binding (members co-signed C, pinned CA roots)
9
verify STARK proof (journal head = C, journal tail = publicIn, image id pinned)
no private-read API, D never accessed
10
commit attestation (status attested, submittedBy, proofDigest)
11
Attest OK
Data owner (member)
ZK prover (off-chain)
Data chaincode CC_D (members)
Members Org1 and Org2
Attestation client (non-member)
Verification chaincode CC_V (non-members)
Ledger